Harrison Henley

Software Engineer

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Summary

Graphics Programmer, learned in C++, with specialisation on low-level game engine development programming used for researching graphical techniques and practical console development. Seeking to utilise skill set to improve graphics within games for the love of high-level rendering and unique graphic techniques.

Education

MCOMP Computer Science for Games
September 2022 – May 2026

Sheffield Hallam University

An Integrated Master's Degree, dedicated to C++ and Graphics Rendering for Game Engine Development. Extensive development in graphics and even published a game to the PlayStation 5.

BTEC Games Design
September 2020 – May 2022

Barnsley College | Merit

Developed an understanding of games development and design.

Roles

Student Representative
September 2023 – May 2026

Sheffield Hallam University

Held a point of contact between students and tutors, giving them assistance where necessary and providing opportunities to improve social skills.

Projects

Raytracing

Used the tutorial NVIDIA released for the Khronos Vulkan Extension to learn how raytracing acceleration structures worked and went through 'Ray Tracing in One Weekend' to learn about the raw raytracing mechanics.

Cascaded / Shadow Maps

Developed a Vulkan application to learn about cascaded shadow maps and consequently, shadow mapping. I learnt my shader code and mathematics from Frank Luna's DirectX12 book and then implemented it for GLSL shader code.

Compute Octree Shaders for Culling (Dissertation)

Learnt about Compute Shaders from the online Vulkan tutorial and implemented them to divide up a scene into multiple groups of areas and if none of the points of the area were visible, it would cull all the vertices belonging to that area. Therefore, speeding up per frame rendering time significantly.

Base Form (PS5 Game Demo – University Work)

A game developed on a proprietary engine, where I had a hand in its creation and development. My main contribution to Base Form, was the implementation of the 'recastnavigation' library for AI pathfinding. Further implementing 3 different enemy types for our game.

Knowledge	Skills
C, C++, C#	Communication
Low Level APIs (Vulkan & DirectX12)	Problem Solving
SDK (PlayStation 5)	Fast Learner
HLSL / GLSL / SPIR-V	Task / Time Keeping – Jira / Trello
RenderDoc, NVIDIA Nsight, RazorGPU	Programming Architecture
Cmake / Lua	Git - GitHub / Git Bash